

The Future of MAID Storage

Jo Maitland

In the five years since its introduction, MAID-based storage deployments can be measured in the hundreds, rather than thousands.

MAID is a storage technology that packs a large number of disc drives into a single array and spins only those that are being used at any given time. This significantly reduces the power consumption of the system, and according to vendors that sell these products, also prolongs the life of the drives. For customers, it means the ability to keep much more data online instead of carting it off to tape. The acronym was coined in the late 1990s at the University of Colorado in Boulder, where students were working on a research project to build high-density online storage that consumed less power. COPAN Systems, located in Longmont, Colorado, productised the research and was the first company to ship a system, in February 2004. Since then, COPAN has installed 300 or so systems worldwide, and a handful of other vendors have joined the market.

Who's Using It And Why?

MAID storage is designed for infrequently accessed data, sometimes referred to as persistent data, where performance (IOPS or bandwidth) is not a big concern. With better response times as compared to access times in magnetic tapes, MAID storage has made some headway in the backup market, where customers are eager to increase service levels and improve the reliability of backups. If you want disc-based backup and have power or cooling constraints in the data center, MAID storage is a killer product. However, most IT shops have retained the traditional tools. Financial services companies, Web 2.0 firms, and government agencies have been early adopters, because they keep large quantities of data online and are often located in major cities short on power. Chicago Mercantile Exchange, Credit Suisse, MySpace.com, and the U.S. Department of Defense are all MAID users.

Drawbacks To Adoption

If you have even considered using MAID, chances are that you ruled it out for a handful of reasons. Limited ways to consume the technology, performance concerns, a lack of tools to identify what data to place on MAID storage,

MAID is expected to be a part of the long-term it strategy of organisations because it has the highest density, lowest power consumption, and longest drive life of any disk-based technology.

At A Glance

Massive array of idle discs (MAID) storage, a technology that radically cuts down on the power consumption of disk drives by spinning them down or off when not in use, has barely made a dent in the market. Despite the hype around green IT, there are several reasons for the slow adoption, in particular the requirement to trade performance for more energy efficiency. This is changing with some interesting shifts in the technology that will likely make MAID storage more appetizing for IT shops over the next 12 months.

and short refresh cycles have all been major obstacles to most firms' implementations:

Limited ways to use the technology. To date, the popular way to access MAID storage has been via a VTL interface in conjunction with a backup application. This has meant a lot of work on your part to figure out how to integrate these systems into an existing backup environment. Furthermore, tape libraries and backup environments are long-term investments where change is unwelcome.

Performance concerns (and myths). There has been a lot of fear, uncertainty, and doubt (FUD) about MAID storage that has obscured some of the real concerns about the technology. Some of the myths floating around have included suggestions that power savings from drives being idle are eaten up by the additional power required to spin the drives back to life. That turns out to be nonsense. And similarly, so is the claim that hard drives fail more often when they are spun on and off. COPAN argues that its software proactively monitors and manages drive health by periodically exercising all disks and detecting potential drive failures before they occur. Better management of the drives actually prolongs the mean time between failures, the company claims.

But there are some less well understood issues around the use of MAID given its performance characteristics. For example, COPAN Systems packs 896 1TB SATA drives into a single cabinet, spinning a maximum of 25 percent of them at any one time and queuing requests as they come in.

In the event that data is requested from a logical unit number (LUN) on a RAID group